Laboratory Exercise 02

Topics

1. Variables in Python
    a. Variable types
    b. Basic arithmetic operators
    c. String manipulation

Discussion

A. What is the relationship between assigned values, variables, and data types?
B. Give an example of operations that can be carried out on variables with assigned values corresponding to different data types.
C. Let's consider the "`+`" operator. What happens if it is used between...
    a. Two variables with values of type `int`;
    b. One variable with a value of type `int` and one with a value of type `float`;
    c. Two variables with values of type `str`;
    d. One variable with a value of type `int` and one with a value of type `str`.

Exercises (in red = suggested ones to complete during the lab)

Part 1 - Arithmetic operations

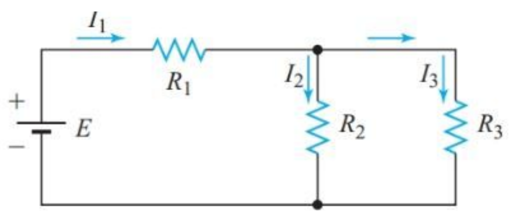
***Goal**: For each of the following exercises, write a Python program that responds to the above requests. Complete at least two exercises during the lab session, and the remaining ones at home.*

[SUGGESTED] 02.1.1 Two numbers. Write a program that stores two integers in two constants defined in code, and then displays them:

A. The sum;
B. The difference;
C. The product;
D. The average value;
E. The distance (i.e. the absolute value of the difference);
F. The maximum value (i.e. the largest of the two);
G. The minimum value (i.e., the smallest of the two).

Tip: Use the `max()` and `min()` functions defined in Python. They take a sequence of comma-separated values as input and return the maximum and minimum values of the sequence, respectively (for example, `max(10, 5)` returns `10`). [P2.4]

02.1.2 Resistances. Consider the following circuit.



Write a program that, starting from the resistances of the three resistors, input by the user, calculates the total resistance, using Ohm's law. [P2.38]

02.1.3 Digits. Write a program that stores a five-digit positive integer (i.e. between 10000 and 99999) in a constant, and displays the individual digits of which it is composed. For example, having the number `16384`, the program should display, on separate lines: `1 6 3 8 4`. [P2.16]

02.1.4 Hybrid car. Write pseudocode and the associated Python program that helps a person decide whether or not to buy a hybrid car. Program inputs should be:

I. the cost of a new car;
II. the estimate of kilometers traveled in a year;
III. The estimation of the cost of fuel;
IV. Efficiency in kilometers per liter;
V. The estimate of the resale value of the used car after 5 years.

Calculate the total cost of ownership of the car for 5 years (for simplicity, do not take into account the cost of financing). To provide input to the program, search the web for realistic prices and fuel consumption for two alternatives for buying a new car in the same price range: a hybrid model and a gasoline model. Run the program twice to compare the two alternatives, considering the current price of fuel and the forecast of driving 30,000 kilometers per year. [P2.10]

[SUGGESTED] 02.1.5 Electric force. According to Coulomb's law, the electric force (expressed in Newtons) between two charged particles with charge, respectively, *Q1* and *Q2*, separated by a distance *r*, is

$$F = \frac{Q_1 \, Q_2}{4 \, \pi \, \varepsilon \, r^2}$$

where $\varepsilon = 8.854 \times 10^{-12}$ Farad/meter. Write a program that calculates and shows on screen the relative force of a pair of charged particles, allowing the user to choose the values *Q1*, *Q2* (in Coulomb) and *r* (in meters). [P2.43]

Part 2 – String Manipulation

Goal*: For each of the following exercises, write a Python program that responds to the above requests. Complete* <u>*at least two*</u> *exercises during the exercise, and the remaining ones at home.*

[SUGGESTED] 02.2.1 Characters. Write a program that stores a string in a variable and, starting from that variable, displays the first three characters of the string, followed by three dots, again followed by the last three characters. For example, if the string is initialized to the value "`Mississippi`", the program should display "`Mis... ppi`".

IMPORTANT: The program should run for any string, of **<u>any length</u>**. What happens if the string is shorter than 6 characters? What if it's shorter than 3 characters? [P2.22]

[SUGGESTED] 02.2.2 Telephone number. The following pseudocode describes how to transform a string containing a ten-digit phone number (such as "`4155551212`") into a more easily readable string, formatted in the U.S. style, such as "`(415) 555-1212`":

I. Take the string consisting of the first three characters and surround it with parentheses (this is the prefix, called "area code");
II. Obtain the final number by concatenating the prefix with the string containing the next three characters, a hyphen, and the string consisting of the last four characters

Translate this pseudocode into a Python program that stores a 10-digit phone number in a string, and then displays it in the format just described. [P2.33]

02.2.3 Alignment. Format the output of Exercise **02.1.1** so that descriptions and numbers are vertically aligned, for example:

```
Sum:        45
Difference: -5
...         ...
```

[P2.5]

02.2.4 Emoji. According to data collected by the Unicode Consortium[1], the non-profit organization responsible for digitizing the world's languages, "tears of joy" (😂) account for more than 5% of all emojis used (the only other character that comes close to it is the ❤️). The top ten emojis used worldwide are: 🙏👍😂😘❤️😍🤣🥰😭😊 .
When you exchange messages on Telegram (or on your favorite messaging app), what are the three emojis that you personally use most frequently? Using the Unicode encoding information collected here[2], write a program that shows on the screen, for each of these three emojis:

I. the position in the ranking;
II. the Unicode number;
III. the Unicode Name;
IV. the emoji itself.

Format the output so that the information for each of the three emojis is collected on one line, and the different fields are vertically aligned.

02.2.5 Enrolments. The student ID number at a university consists of two parts: a letter and a sequence of numbers. Write a program that, starting from two ID numbers, shows them on the screen in ascending order based on the numerical part only. Tip: Use the built-in functions of the Python language.

[1] https://home.unicode.org/emoji/emoji-frequency/
[2] https://unicode-table.com/